
School of Mathematics and Statistics

MT5864 Advanced Group Theory

Problem Sheet II: Group Actions

1. Let G be a group and let Γ and Δ be sets such that G acts on Γ and on Δ . Define

$$(\gamma, \delta)^x = (\gamma^x, \delta^x)$$

for all $\gamma \in \Gamma$, $\delta \in \Delta$ and $x \in G$. Verify that this is an action of G on the set $\Gamma \times \Delta$.

Show that the stabilizer of the pair (γ, δ) in this action equals the intersection of the stabilizers G_γ and G_δ (these being the stabilizers under the actions of G on Γ and Δ , respectively).

If G acts transitively on the non-empty set Ω , show that

$$\{(\omega, \omega) \mid \omega \in \Omega\}$$

is an orbit of G on $\Omega \times \Omega$. Deduce that G acts transitively on $\Omega \times \Omega$ if and only if $|\Omega| = 1$.

2. (a) By definition, the symmetric group S_n acts on the set $\Omega = \{1, 2, \dots, n\}$. How many orbits does S_n have in its induced action on $\Omega \times \Omega$?
 (b) Repeat part (a) with the action of the alternating group A_n on Ω .

3. Let G be a group and $\rho: G \rightarrow \text{Sym}(\Omega)$ be a permutation representation. Show that this determines an action of G on Ω by the formula

$$\omega^x = \omega^{x\rho} \quad \text{for } \omega \in \Omega \text{ and } x \in G.$$

[On the right hand side, $\omega^{x\rho}$ denotes the image of ω under the permutation $x\rho$.]

Deduce that there is a one-one correspondence between actions of a group G on a set Ω and the permutation representations $G \rightarrow \text{Sym}(\Omega)$.

4. Let G be a group that acts transitively on a set Ω , let $\omega \in \Omega$ and let H be a subgroup of G . Show that H is transitive on Ω if and only if $G = G_\omega H$.
5. (a) Let $i_1, i_2, \dots, i_k$ be distinct points in $\Omega = \{1, 2, \dots, n\}$ and let σ be a permutation in S_n . By considering the effect on various points in Ω , or otherwise, show that

$$\sigma^{-1}(i_1 \ i_2 \ \dots \ i_k)\sigma = (i_1\sigma \ i_2\sigma \ \dots \ i_k\sigma).$$

[To reduce the number sub/superscripts, in this formula we are writing $i\sigma$ for the image of a point i under the permutation σ .]

Deduce that two permutations in S_n are conjugate if and only if they have the same cycle structure.

- (b) Give a list of representatives for the conjugacy classes in S_5 . How many elements are there in each conjugacy class? Hence calculate the order of and generators for the centralizers of these representatives.
6. The symmetric group S_n of degree n acts in its usual definition on the set $\Omega = \{1, 2, \dots, n\}$. In this question, we write $i\sigma$ for the image of $i \in \Omega$ under a permutation σ to reduce the need for double sub/superscripts.

We now define an action of S_n on another set.

Let X be any set and define

$$\Delta = X^{(n)} = \{ (x_1, x_2, \dots, x_n) \mid x_1, x_2, \dots, x_n \in X \}.$$

If σ is a permutation in the symmetric group S_n and $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \Delta$, define

$$\mathbf{x}^\sigma = (x_{1\sigma^{-1}}, x_{2\sigma^{-1}}, \dots, x_{n\sigma^{-1}}).$$

[Consequently, if a permutation σ moves i to j (that is, $i\sigma = j$) then applying σ to a sequence $\mathbf{x} = (x_1, x_2, \dots, x_n)$ moves the content of position i in $\mathbf{x}$ into position j in $\mathbf{x}^\sigma$. This is why the inverse σ^{-1} appears in the formula.]

Show that this defines an action of S_n on Δ .

7. Let F be a field and $V = F^n$ be the vector space of dimension n over F consisting of row vectors.
- (a) Let $\mathcal{B} = \{e_1, e_2, \dots, e_n\}$ denote the standard basis for the vector space V . Let $k, \ell \in \{1, 2, \dots, n\}$ with $k \neq \ell$ and α be a non-zero scalar in F . Show that the map

$$g: e_k \mapsto e_k + \alpha e_\ell, \quad e_i \mapsto e_i \quad \text{for } i \neq k$$

is an element of $\text{GL}_n(F)$. [A linear map of this form is called a *transvection*.]

- (b) Let $h = [h_{ij}]$ be a matrix in $\text{GL}_n(F)$ and suppose that h commutes with the above transvection g . Show that $h_{\ell j} = 0$ for $j \neq \ell$ and $h_{kk} = h_{\ell\ell}$.
- (c) Deduce that the centre of $\text{GL}_n(F)$ consists of the scalar matrices:

$$\text{Z}(\text{GL}_n(F)) = \{ \lambda I \mid \lambda \in F^* \}$$

(where I denotes the identity matrix).

- (d) Let Ω be the set of 1-dimensional subspaces of V and $\rho: \text{GL}_n(F) \rightarrow \text{Sym}(\Omega)$ be the permutation representations determined by the natural action of $\text{GL}_n(F)$ on Ω . Show that the kernel of ρ equals the set of scalar matrices.

[Combining the last two parts tells us that the group of permutations induced by $\text{GL}_n(F)$ on the set Ω of 1-dimensional subspaces is isomorphic to the quotient $\text{GL}_n(F)/\text{Z}(\text{GL}_n(F))$. We call this the *projective general linear group* $\text{PGL}_n(F)$.]

- (e) Show that the centre of the special linear group $\text{SL}_n(F)$ consists of all scalar matrices of determinant 1. [Hint: What is the determinant of a transvection?]
- (f) Deduce that the *projective special linear group* $\text{PSL}_n(F) = \text{SL}_n(F)/\text{Z}(\text{SL}_n(F))$ is isomorphic to a subgroup of the projective general linear group $\text{PGL}_n(F)$.

8. Let F be a finite field of order $q = p^k$, where p is prime, and n be a positive integer.
- Determine the order of the general linear group $\text{GL}_n(F)$.
[Hint: A matrix is invertible if and only if its rows are linearly independent.]
 - Find an example of a Sylow p -subgroup of $\text{GL}_n(F)$ (where p is the characteristic of the field F).
 - Show that $\text{GL}_n(F)$ acts on the set $V \setminus \{\mathbf{0}\}$ of non-zero vectors in the vector space $V = F^n$.
 - By considering the action on the set of non-zero vectors, or otherwise, show that $\text{GL}_2(\mathbb{F}_2)$ is isomorphic to the symmetric group S_3 of degree 3.
9. Let G be a group and let G act on itself by conjugation. Let $\tau: G \rightarrow \text{Sym}(G)$ be the associated permutation representation.
- What is the image of τ ?
 - What is the kernel of τ ?
 - Deduce a useful isomorphism result from these observations.
10. Let G be a group and H be a subgroup of G . Show that the normalizer $N_G(H)$ of H is the largest subgroup of G in which H is a normal subgroup.
[By *largest*, we mean that if L is any subgroup of G in which H is normal, then $L \leq N_G(H)$. So you should check that (i) $H \trianglelefteq N_G(H)$ and (ii) if $H \trianglelefteq L$ then $L \leq N_G(H)$.]
11. (a) Let G be a group and consider the right regular action of G on itself. (Recall this is given by $(g, x) \mapsto gx$.)
Show that G acts transitively on itself by this formula. If $g \in G$, determine the stabilizer of g under this action.
- (b) Let G be a group and H be a subgroup of G . Let H act on G via the induced action obtained from the right regular action; that is,

$$\begin{aligned} G \times H &\rightarrow G \\ (g, x) &\mapsto gx. \end{aligned}$$

Show this is an action of H on G . Determine the orbits under this action of H . What theorem from group theory can you deduce from this action?

- (c) Let G be any group. Define $*$ by the formula

$$g * x = x^{-1}g$$

and view this as applying x to the element g . Show that this defines an action of G on itself. (This is the *left regular action* of G on itself.)

Is this action transitive? If $g \in G$, determine the stabilizer of g under this action.

Let H be a subgroup of G and consider the induced action of H on G obtained from the left regular action:

$$\begin{aligned} G \times H &\rightarrow G \\ (g, x) &\mapsto g * x = x^{-1}g. \end{aligned}$$

Determine the orbits under this action of H . What theorem from group theory can you deduce?

12. If H is a subgroup of G of index n , show that the index of the core of H in G divides $n!$.
[Hint: Use the action on cosets.]

13. Let G be a group and suppose that it has *equivalent* actions on the sets Ω and Ω' . Let $\rho: G \rightarrow \text{Sym}(\Omega)$ and $\rho': G \rightarrow \text{Sym}(\Omega')$ be the associated permutation representations.

- (a) Show that $\ker \rho = \ker \rho'$.
(b) Show that the permutation groups $G\rho$ and $G\rho'$ are permutation isomorphic.

14. Let $G \leq \text{Sym}(\Omega)$ and $H \leq \text{Sym}(\Omega')$. Suppose that G and H are permutation isomorphic via the bijection $\phi: \Omega \rightarrow \Omega'$ and the group isomorphism $\theta: G \rightarrow H$. Define

$$\omega^x = \omega^{x\theta}$$

for each $\omega \in \Omega'$ and $x \in G$. Show that this is an action of G on Ω' and that it is equivalent to the action of G on Ω .

15. Let G be a group. Show that equivalence (as defined in the notes) is an equivalence relation on the collection of actions of G on sets.

[Show that equivalence of actions is reflexive, symmetric and transitive as a relation.]

16. Show that there is no simple group of order equal to each of the following numbers:

- (i) 30; (ii) 48; (iii) 54; (iv) 66; (v) 72;
(vi) 84; (vii) 104; (viii) 132; (ix) 150; (x) 392.

[Note: Some of these can be solved using Sylow's Theorem and methods from *MT4003* alone, while others require the technology of group actions. They are not necessarily in increasing order of difficulty!]

17. Let G be a simple group of order 60.

- (a) Show that G has no proper subgroup of index less than 5. Show that if G has a subgroup of index 5, then $G \cong A_5$.

[Hint: If H is a subgroup of index k , let G act on the set of cosets to produce a permutation representation. What do we know about the kernel?]

- (b) Let S and T be distinct Sylow 2-subgroups of G . Show that if $x \in S \cap T$ then $|C_G(x)| \geq 12$. Deduce that $S \cap T = 1$. [Hint: What do you know about S and T from their orders?]

- (c) Deduce that G has at most five Sylow 2-subgroups and hence that indeed $G \cong A_5$.

Thus we have shown that there is a unique simple group of order 60 up to isomorphism.

18. Let p be a prime number, P be a finite p -group and let P act on a finite set Ω . Define

$$\text{Fix}_P(\Omega) = \{ \omega \in \Omega \mid \omega^x = \omega \text{ for all } x \in P \},$$

the set of fixed-points in the action. By considering the orbits for the action, show that

$$|\text{Fix}_P(\Omega)| \equiv |\Omega| \pmod{p}.$$

19. Let G be a finite p -group and N be a non-trivial normal subgroup of G . Show that $Z(G) \cap N \neq \mathbf{1}$.

[Hint: Let G act by conjugation on N .]